

LayoutLab AR

Make invisible Java layout rules visible.

Immersive Digital Teaching Aid | Grup Nelang PMU | Politeknik Mukah
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01 BACKGROUND / PROBLEM

Novice diploma students in DFP50463 Java Based Application Development at Politeknik Mukah struggle to predict resizing, hidden components and nested panels. Slides explain rules; GUI builders can conceal them. Learners need to see, test and repair the behaviour. [1]

02 INNOVATION OBJECTIVES

- **Construct** a BorderLayout prototype in tracked AR.
- **Test and diagnose** resizing and region collisions.
- **Repair and explain** the structure using verifiable Java evidence.

NOSS IT-010-3:2016-C01 · Application Prototype Development

04 DEVELOPMENT PROCESS

- 1 **Identify** Initial student feedback (n=11) established the need.
- 2 **Design** Map construct–test–diagnose–repair to NOSS tasks.
- 3 **Develop** React, Three.js and MindAR; shared Java generator.
- 4 **Test** Automated checks, follow-up users (n=6), device acceptance.
- 5 **Refine** Improve onboarding, touch targets and resizing feedback.

07 VALIDATION & EVIDENCE

81.8%

valued visual clarification
Initial feedback · n=11

6/6

reported layout insights
Follow-up feedback · n=6

AR acceptance: iPhone 13 demonstration documented tracking, touch, audio, 3/3 missions and Java evidence.

Formative self-reports; no controlled AR learning gain measured. Source: project extended abstract.

08 EXPECTED IMPACT

Learning: practise diagnosis and explain repairs. **Institution:** repeatable browser demonstrations. **Industry/community:** prototype skills. **Commercialisation:** future challenge packs. **TVET:** competency-based digital practice.

03 THE INNOVATION

A browser-based AR teaching aid that turns a virtual JFrame into a tracked model, touch surface and assessed task. Visual, audio and code feedback connect each action to a Java rule.

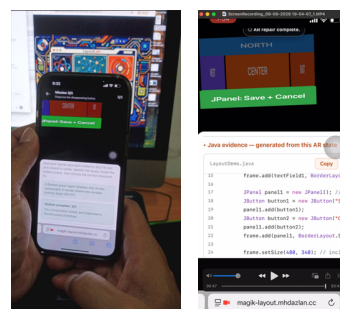


Figure 1. Actual AR use and generated Java after structural repair.

CONSTRUCT

Place a title in NORTH.

TEST

Predict how CENTER resizes.

REPAIR

Nest Save + Cancel in one JPanel; inspect the Java.

05 VALUE & ADVANTAGES

Novelty. AR interaction links behaviour, repair and code.

Usefulness. Makes layout rules observable and assessable.

Replicability. Reusable tasks for other courses and institutions.

Cost effectiveness. Browser delivery avoids native installation.

Sustainability. Reusable engine and tests support maintenance.

Scalability. Further NOSS modules and institutional packs.

06 AI USE & ETHICS

OpenAI Codex assisted development and writing. Authors verify outputs against code, tests and device evidence. AR scores and assessed Java come from deterministic rules; generative AI does not decide correctness.

09 CONCLUSION

LayoutLab AR connects spatial action, tested behaviour, structural repair and verifiable code. The prototype is ready for demonstration. Next: a classroom pilot measuring completion time, errors and learning performance.

10 REFERENCES & SUPPORT

[1] Qian, Y., & Lehman, J. (2017). Students' misconceptions and other difficulties in introductory programming: A literature review. *ACM Transactions on Computing Education*, 18(1), Article 1. <https://doi.org/10.1145/3077618>

[2] Keuning, H., Jeuring, J., & Heeren, B. (2018). A systematic literature review of automated feedback generation for programming exercises. *ACM Transactions on Computing Education*, 19(1), Article 3. <https://doi.org/10.1145/3231711>

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